



Student Name: _____

Teacher: _____

Sydney Technical High School

September
2024

Year 11
PRELIMINARY
EXAMINATION

Mathematics Extension 1

General Instructions

- Working Time – 1 hour and 30 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless work or illegible writing
- **Begin each question on a new page**

Total marks:
70

Section I — 7 marks

- Attempt Questions 1 - 7
- Allow about 10 minutes for this part

Section II — 63 marks

- Attempt Questions 8 - 14
- Allow about 80 minutes for this section.

SECTION 1

7 marks

Attempt Questions 1 – 7

Allow about 10 minutes for this section.

Use the multiple – choice answer sheet for questions 1 – 7

1. The arrangements of the letters of the word EXCELLENT is

- A $9!$
 - B $\frac{9!}{3!}$
 - C $\frac{9!}{5!}$
 - D $\frac{9!}{2 \times 3!}$
-

2. How many solutions does the equation $(2\sin x - 1)(\tan 3x - 1) = 0$ have in the domain $0 \leq x \leq 2\pi$?

- A 4
 - B 6
 - C 8
 - D 1
-

3. The numbers α, β , and γ satisfy the equations

$$\alpha + \beta + \gamma = 3$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1 \text{ and}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 2$$

α, β , and γ are the roots of;

- A $x^3 + 3x^2 - x + 2 = 0$
 - B $x^3 - 3x^2 + x - 2 = 0$
 - C $x^3 + 3x^2 - 4x + 2 = 0$
 - D $x^3 - 3x^2 + 4x - 2 = 0$
-

4. The graph of $y = kx - 4$ intersects the graph of $y = x^2 + 2x$ at two distinct points for

- A $k < -2$ or $k > 6$
 - B $-2 \leq k \leq 6$
 - C $6 - 2\sqrt{3} \leq k \leq 6 + 2\sqrt{3}$
 - D $k = -2$
-

5. Two events, A and B, are independent, where $P(B) = 2P(A)$ and $P(A \cup B) = 0.52$. $P(A)$ is equal to

- A 0.2
 - B 0.3
 - C 0.4
 - D 0.5
-

6. Whenever n integers are written on a whiteboard, at least six of them leave the same remainder when divided by 3.

What is the smallest possible value of n ?

- A 3
 - B 7
 - C 15
 - D 16
-

7. The parametric equation of the point T is $x = \frac{pq}{p+q}$, $y = \frac{1}{p+q}$ where p and q are parameters and $(p - q)^2 = 4$ is:

- A $x = \frac{1}{y} - 4y$
 - B $1 - 4xy = 4y^2$
 - C $1 - 2xy = 4y^2$
 - D $x^2 + y^2 - xy = 4$
-

SECTION II

63 marks

Attempt Questions 8 – 14

Allow about 80 minutes for this section.

Question 8 (9 marks) Start this question at the top of a NEW Page

- a. Solve the inequality $|2x - 1| > 5$ 2
- b. The polynomial $P(x) = 2x^3 - bx^2 - cx + 10$ is divisible by $(x - 1)(x + 2)$.
Find the value of b and c . 3
- c. Find the value of the constant term in the expansion $\left(3x - \frac{2}{x}\right)^{12}$ 2
- d. How many arrangements are there if 8 Royals are to be seated around a round table, given that William and Harry cannot be seated next to each other, or directly opposite each other. 2

Question 9 (9 marks) Start this question at the top of a NEW Page

- a. i. Solve $2x^2 - 9x + 4 > 0$ 2
ii. Hence or otherwise find the value of m for which $2x^2 - mx + 4 \geq 0$ for all x 2
- b. Find the general solution to $2\sin x + \sqrt{3} = 0$ 2
- c. i. Sketch the graph of $y = x^2 + 4x$ clearly indicating any intercepts and turning point. 1
ii. Hence, sketch the graph of $y = \frac{1}{x^2 + 4x}$ 2

Question 10 (9 marks) Start this question at the top of a NEW Page

- a. The graphs $y = 2x + 1$ and $y = 3^x$ intersect at $A(1,3)$ and $B(x,y)$
- i. State the co-ordinates of B **1**
- ii. Solve the inequality $3^x > 2x + 1$ **2**
- b. Given $\log_a b = 3.4$ and $\log_b c = 4.7$ find the value of $\log_a c$ **2**
- c. A population of insects is measured using the rule $P = 1000 + (4.25)^{1.4t}$ where t is measured in days.
- i. State the initial population of insects **1**
- ii. State the number of insects present after 5 days? **1**
- iii. How long does it take the population to reach 1 million?
Give your answer correct to the nearest hour. **2**

Question 11 (9 marks) Start this question at the top of a NEW Page

- a. i. $P(x) = x^4 - 5x^3 + 6x^2 + 4x - 8$ has a root of multiplicity 3.
Find this root and hence express $P(x)$ as a product of its factors. **3**
- ii. Sketch the curve $y = P(x)$,
given the curve passes through the point $(-0.25, -8.54297)$ **2**
- iii. Hence, on separate axes, sketch each of the following,
- $\alpha.$ $y = -f|x + 2|$ **2**
- $\beta.$ $y^2 = f(x)$ **2**

Question 12 (9 marks) Start this question at the top of a NEW Page

- a. For some function $y = h(x)$, where $h(x) = f(x).g(x)$, it is known that

$$g(1) = 2, f(1) = 4, g'(x) = -1 \text{ \& } f'(x) = 3.$$

Find the equation of the tangent to the curve $y = h(x)$, at the point where $x = 1$. **3**

- b. Consider the function,

$$f(x) = 2 + \frac{1}{x-2}$$

- i. Give the equation of the vertical asymptote for $y = f(x)$ **1**
- ii. By sketching $g(x) = x$ and $y = f(x)$ on the same axes, showing their points of intersection, sketch the function $y = f(x).g(x)$ **3**
- iii. Find the values of x for which the tangents to the curve $y = f(x).g(x)$ are horizontal **2**

Question 13 (9 marks) Start this question at the top of a NEW Page

- a. Solve $\frac{x^2-2x}{1-x} > 1$ 3
- b. Given $\sin(A+B) = \sin A \cos B + \sin B \cos A$.
- i. Show that the value of $\sin\left(\frac{\pi}{12}\right)$ is $\frac{\sqrt{3}-1}{2\sqrt{2}}$ 2
- ii. Hence, show that the exact value of $\cos\left(\frac{\pi}{12}\right)$ can be expressed as $\frac{\sqrt{2+\sqrt{3}}}{2}$ 2
- c. A committee of 12 is to be created from 16 Republicans and 14 Democrats.
What is the probability that the majority of the committee will be from one party? 2

Question 14 (9 marks) Start this question at the top of a NEW Page

- a. It is known that the function $y = \log_{10}(x+2) - \log_{10}(x-1)$ has an inverse function in the domain $x > 1$
Find the equation for $y = f^{-1}(x)$ and state the domain for this inverse. 3
- b. In the expansion of $(a+x)^2(3-2x)^7$, the coefficient of x^8 is -320 .
Find the value of a . 3
- c. By considering the expansion of $(a+bx)^{10}$ with suitable values of a and b ,
Show that $1 + {}^{10}C_2 3^2 + {}^{10}C_4 3^4 + {}^{10}C_6 3^6 + {}^{10}C_8 3^8 + 3^{10} = 2^9[2^{10}+1]$ 3

END OF ASSESSMENT TASK

Year 11 - Ext 1 YEARLY (TASK 3) 2024

1. 9 letters $3 \times E$ $2 \times L$

$$\therefore \frac{9!}{3! \times 2!} \quad \text{OR} \quad \frac{9!}{2 \times 3!} \quad \textcircled{D}$$

2. $(2\sin x - 1)(\tan 3x - 1) = 0$ $\frac{S}{T} \mid \frac{A}{C}$

$$x = \frac{\pi}{6}, \frac{5\pi}{6} \quad 3x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}, \frac{17\pi}{4}, \frac{21\pi}{4}$$

test as sine is with tan

tan $\tan 3(\pi/6)$ and $\tan 3(5\pi/6)$ are undefined $\therefore 6 \text{ sol}^n$ (B)

3. $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$$1 = 9 - 2(\text{doubles})$$

$$\text{doubles} = 4$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\text{doubles}}{\alpha\beta\gamma} = 2$$

$$\alpha\beta\gamma = 2$$

Now $x^3 - (\text{triples})x^2 + (\text{doubles})x - \text{triples} = 0$

$$x^3 - 3x^2 + 4x - 2 = 0$$

(D)

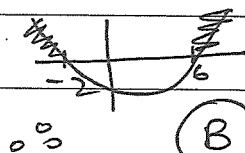
4. $kx - 4 = x^2 + 2x$

$$x^2 + 2x - kx + 4 = 0$$

$$x^2 + (2-k)x + 4 = 0$$

$$\Delta > 0 \quad (2-k)^2 - 4(1)(4) > 0$$

$$(k-6)(k+2) > 0$$



(B)

5. $(p-q)^2 = 4$

$$p^2 - 2pq + q^2 = 4$$

$$(p+q)^2 - 4pq = 4$$

↓

$$\left(\frac{1}{y}\right)^2 - 4\left(\frac{x}{y}\right) = 4$$

$$1 - 4xy = 4y^2$$

$$x = \frac{pq}{p+q}$$

↓

$$(p+q)x = pq$$

$$\therefore \frac{x}{y} = pq$$

$$y = \frac{1}{p+q}$$

↓

$$y(p+q) = 1$$

$$p+q = \frac{1}{y}$$

(B)

$$6. P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A) \quad \text{independent}$$

$$\text{let } P(A) = x$$

$$\therefore P(B) = 2x$$

$$\text{NOW } P(A \cap B) = x \cdot 2x$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.52 = x + 2x - 2x^2 \quad \text{Solve}$$

$$2x^2 - 3x + 0.52 = 0$$

$$2(0.2)^2 - 3(0.2) + 0.52 = 0$$

OR
test sol^{ns}

$\therefore$ (A)

7. $\div$ by 3 gives remainders 0, 1 or 2

at least 6 have the same

ie 5 have the same when $n = 15$ (5×3)

$\therefore$ Pigeonhole 16 are needed to have at least

6 the same (n+1)

$\therefore$ (B)

Question 8

a) $|2x-1| > 5$

$$2x-1 > 5 \quad \text{or} \quad -2x+1 > 5$$

$$2x > 6$$

$$x > 3$$

$$-2x > 4$$

$$x < -2$$

$$\therefore x < -2, x > 3$$

b) $P(x) = 2x^3 - bx^2 - cx + 10$

$$P(1) = 2 - b - c + 10 = 0 \longrightarrow 12 = b + c$$

$$P(-2) = 2(-8) - 4b + 2c + 10 = 0$$

$$-16 - 4b + 2c + 10 = 0$$

$$-8 - 2b + c + 5 = 0 \longrightarrow -2b + c = 3$$

$$\therefore 3b = 9$$

$$b = 3$$

$$c = 9$$

c) $(3x - 2/x)^{12}$

$$\text{general term} = {}^{12}C_k (3x)^{12-k} (-2x^{-1})^k$$

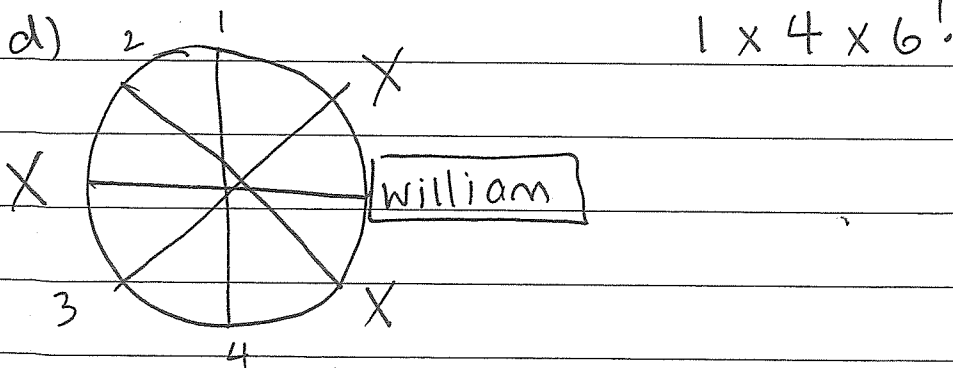
equate to power of zero

$$x^{12-k} \cdot x^{-k} = x^0$$

$$12 - 2k = 0$$

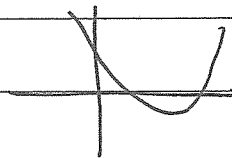
$$k = 6$$

$$\therefore {}^{12}C_6 (3)^6 (-2)^6$$



Question 9

a) i. $2x^2 - 9x + 4 > 0$
 $(2x-1)(x-4) > 0$



$$x < 1/2 \text{ or } x > 4$$

ii. $\Delta < 0$ $m^2 - 4(2)(4) < 0$
 $m^2 - 32 < 0$

$$(m - 4\sqrt{2})(m + 4\sqrt{2}) < 0$$

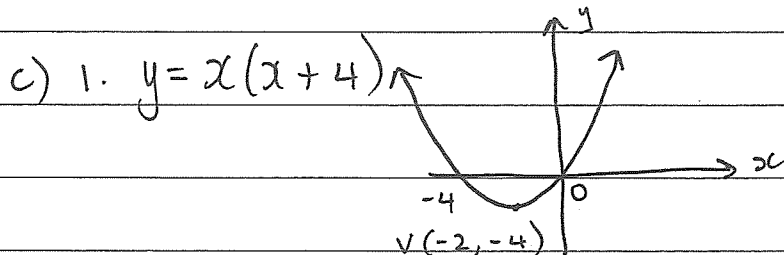
$$-4\sqrt{2} < m < 4\sqrt{2}$$

b)

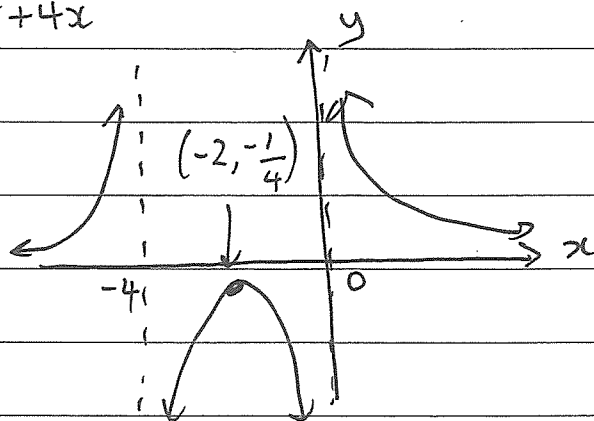
$$\sin x = -\sqrt{3}/2$$

$$x = n\pi + (-1)^n \cdot \left(-\frac{\pi}{3}\right) \quad n \in \mathbb{Z}$$

OR accept: $x = \frac{4\pi}{3} + 2\pi n$, $\frac{5\pi}{3} + 2\pi n$ $n = 0, 1, 2, 3, \dots$
(similar equivalent to)



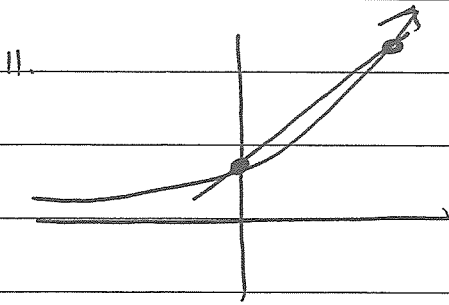
ii. $y = \frac{1}{x^2 + 4x}$



Need T.P
marked
and clear
shape w/
asymptotes
for both marks.

Question 10

a) i. $2x+1 = 3^x$ By inspection $B(0,1)$



$$3^x > 2x+1 \quad \text{exp above line}$$

$$\therefore x < 0 \text{ or } x > 1$$

b) $\log_a c = \frac{\log_b c}{\log_b a}$

$$= 4.7 \div \frac{1}{3.4}$$

$$= 4.7 \times 3.4$$

$$= 15.98$$

take

$$\log_a b = 3.4$$

$$\frac{\log_b b}{\log_b a} = 3.4$$

$$\frac{1}{\log_b a} = 3.4$$

$$\log_b a$$

$$\log_b a = \frac{1}{3.4}$$

c) $P = 1000 + (4.25)^{1.4t}$

i. $t=0 \quad P = 1000 + (4.25)^0$
 $= 1001$

ii. $t=5 \quad P = 1000 + (4.25)^{1.4(5)}$
 $= 38366.945$
 $\approx 38,367 \text{ insects}$

iii. $1000000 = 1000 + (4.25)^{1.4t}$

$$999000 = 4.25^{1.4t}$$

$$\log(999000) = 1.4t \log(4.25)$$

$$t = \frac{\log(999000)}{1.4 \log(4.25)}$$

$$t = 6.8196 \dots \text{ days}$$

$$= 163.672 \dots \text{ hours}$$

$$\approx 164 \text{ hours}$$

Question 11

Need

$$1. P(x) = x^4 - 5x^3 + 6x^2 + 4x - 8$$

1.

$$P'(x) = 4x^3 - 15x^2 + 12x + 4$$

2.

$$P''(x) = 12x^2 - 30x + 12$$

3.

$$\text{Now } P''(x) = 6(2x^2 - 5x + 2) = 0$$

$$6(2x - 1)(x - 2) = 0$$

$$\therefore P''(1/2) \text{ and } P''(2) = 0$$

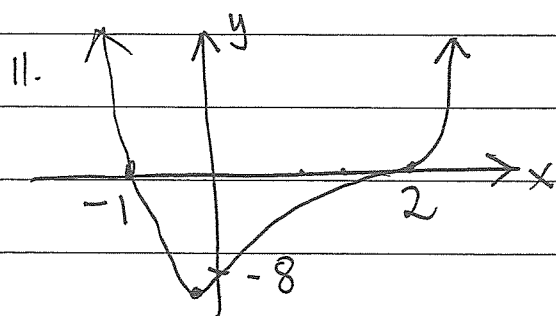
$$\text{test } P(1/2) \neq 0 \quad P(2) = 0$$

$$\therefore \text{ roots are } \alpha, 2, 2, 2 \quad \alpha \beta \gamma \delta = -8$$

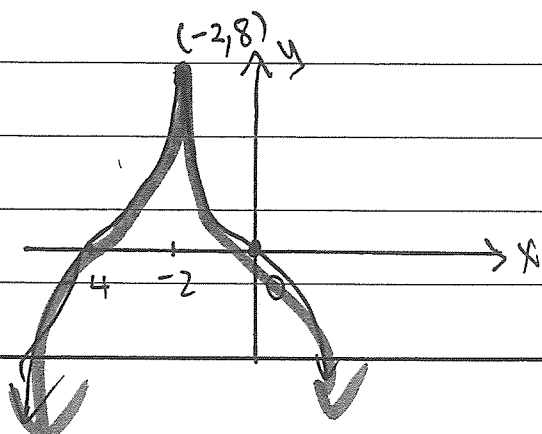
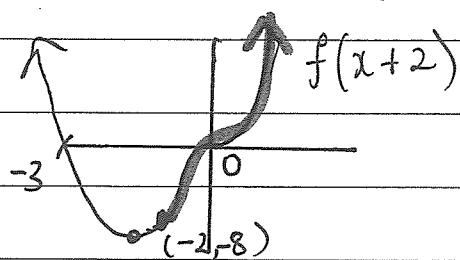
$$2^3 \times \alpha = -8$$

$$\therefore P(x) = (x+1)(x-2)^3 = 0$$

$$\alpha = -1$$

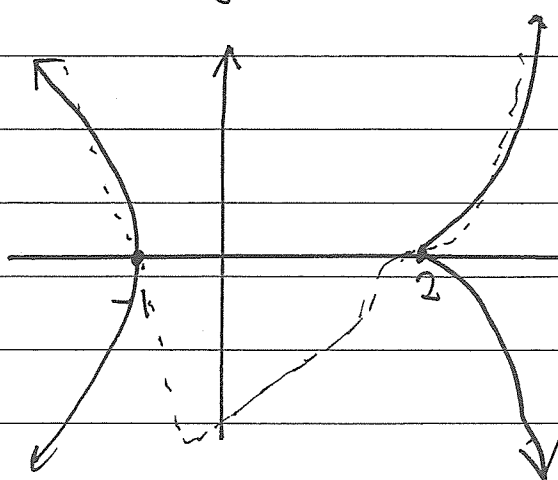


111. (a) $y = -f|x+2|$



(b) $y^2 = f(x)$

$$y = \pm \sqrt{f(x)}$$



Question 12

a) $x=1$ $h(1) = f(1)g(1) = 4 \times 2 = 8 \quad \therefore (1, 8)$

$$h'(x) = f(x) \cdot \underline{g'(x)} + \underline{f'(x)} g(x)$$

$$\therefore h'(1) = f(1) \cdot \underline{g'(1)} + \underline{f'(1)} \cdot g(1)$$

$$= 4 \times \underline{(-1)} + \underline{(3)} \cdot (2)$$

$$= -4 + 6$$

$$= 2$$

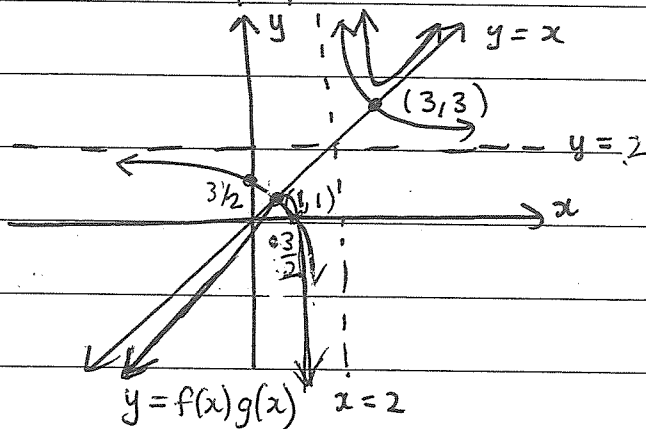
$\therefore$ tangent $y - 8 = 2(x - 1)$

$$y = 2x + 6$$

b) $f(x) = 2 + \frac{1}{x-2}$

i. vertical asymptote $x=2$

ii.



$$x \rightarrow \infty \quad \frac{1}{x-2} \rightarrow 0$$

$$\therefore y \rightarrow 2$$

$$0 = 2 + \frac{1}{x-2}$$

$$-2 = \frac{1}{x-2}$$

$$-2x + 4 = 1$$

$$x = 3/2$$

$$2 + \frac{1}{x-2} = x$$

$$2(x-2) + 1 = x(x-2)$$

$$2x - 4 + 1 = x^2 - 2x$$

$$x^2 - 4x + 3$$

$$(x-3)(x-1) = 0$$

iii) $y = 2x + \frac{x}{x-2}$

$$\frac{dy}{dx} = 2 + \frac{(x-2)(1) - (x)(1)}{(x-2)^2} = 0$$

$$2 + \frac{x-2-x}{(x-2)^2} = -2$$

$$-2 = -2(x^2 - 4x + 4)$$

$$+1 = x^2 - 4x + 4$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 3, x = 1$$

(or) From pts of intersection in part (ii).

Question 13.

a) $x \neq 1$ $\frac{x(x-2)}{1-x} > 1$ $x(1-x)^2$ both sides

$$x(x-2)(1-x) > (1-x)^2$$

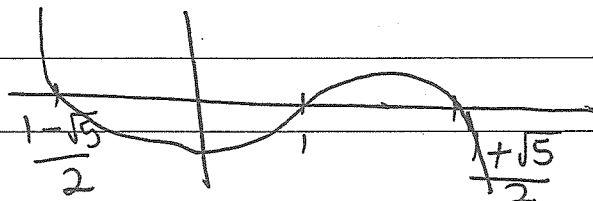
$$x(x-2)(1-x) - (1-x)^2 > 0$$

$$(1-x)[x^2 - 2x - 1 + x] > 0$$

$$(1-x)(x^2 - x - 1) > 0$$

let $x^2 - x - 1 = 0$

$$x = \frac{1 \pm \sqrt{5}}{2}$$



$$\therefore x < \frac{1 - \sqrt{5}}{2}$$

$$1 < x < \frac{1 + \sqrt{5}}{2}$$

b) $\sin\left(\frac{\pi}{4} + \left(-\frac{\pi}{6}\right)\right) = \sin\frac{\pi}{4} \cos\left(-\frac{\pi}{6}\right) + \sin\left(-\frac{\pi}{6}\right) \cos\left(\frac{\pi}{4}\right)$

$$= \sin\frac{\pi}{4} \cos\frac{\pi}{6} - \sin\frac{\pi}{6} \cos\frac{\pi}{4}$$

as $\sin x$ is odd fnc

$\cos x$ is even fnc $= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

ii) $\cos\frac{\pi}{12}$

$$8 = (\sqrt{3} - 1)^2 + x^2$$

$$8 = 4 - 2\sqrt{3} + x^2$$

$$x^2 = 4 + 2\sqrt{3}$$

$$\therefore \cos\frac{\pi}{12} = \frac{\sqrt{4+2\sqrt{3}}}{2\sqrt{2}}$$

$$x = \sqrt{4+2\sqrt{3}}$$

$$= \sqrt{\frac{4+2\sqrt{3}}{8}}$$

$$= \sqrt{\frac{2+\sqrt{3}}{4}} = \frac{\sqrt{2+\sqrt{3}}}{2}$$

$$c) 16R \quad 14D \quad \text{no restrictions} = {}^{30}C_{12}$$

majority is $R > D$ or $D > R$ ie not $R = D$

$$\therefore 1 - \text{Prob}(R = D)$$

$$= 1 - \frac{{}^{16}C_6 \times {}^{14}C_6}{{}^{30}C_{12}} \quad \leftarrow \text{accept here.}$$

$$= 0.721966 \dots$$

Question 14

$$a) y = \log_{10}(x+2) - \log_{10}(x-1) \quad x > 1$$

creating inverse $x = \log_{10}(y+2) - \log_{10}(y-1)$

$$x = \log_{10}\left(\frac{y+2}{y-1}\right)$$

$$10^x = \frac{y+2}{y-1}$$

$$10^x(y-1) = y+2$$

$$10^x(y) - 10^x = y+2$$

$$10^x(y) - y = 2 + 10^x$$

$$y(10^x - 1) = 2 + 10^x$$

$$y = \frac{2 + 10^x}{10^x - 1}$$

$$f^{-1}(x) = \frac{2 + 10^x}{10^x - 1}$$

for: $f^{-1}(x)$

Domain: $x > 0$

Range: $y > 1$

$$b) (a+x)^2 (3-2x)^7$$

$$(a^2 + 2ax + x^2) \left[{}^7C_0(3)^7 - {}^7C_1 3^6(2x)^6 + \dots + {}^7C_6 3(-2x)^6 + {}^7C_7(-2x)^7 \right]$$

$$\text{coeff of } x^8 \left\{ \begin{array}{l} \text{consider terms involving:} \\ x \times x^7 + x^2 \times x^6 \end{array} \right\}$$

$$-320 = 2a {}^7C_7 (-2)^7 + 1 \times {}^7C_6 (3)(-2)^6$$

$$-320 = 2a \times 1 \times (-2)^7 + 1 \times 7 \times 3 \times (-2)^6$$

$$-320 = -256a + 1344$$

$$256a = 1664$$

$$a = 6.5.$$

$$c) (a+bx)^{10} \text{ use } (1+3x)^{10}$$

$$(1+3x)^{10} = {}^{10}C_0 + {}^{10}C_1(3x) + {}^{10}C_2(3x)^2 + {}^{10}C_3(3x)^3 + \dots + {}^{10}C_9(3x)^9 + {}^{10}C_{10}(3x)^{10}$$

$$\text{let } x=1 \text{ into (1)}$$

$$4^{10} = {}^{10}C_0 + {}^{10}C_1(3) + {}^{10}C_2(3)^2 + {}^{10}C_3(3)^3 + \dots + {}^{10}C_9(3)^9 + {}^{10}C_{10}(3)^{10} \quad \text{--- (2)}$$

$$\text{let } x=-1 \text{ into (1)}$$

$$\begin{aligned} (-2)^{10} &= {}^{10}C_0 + {}^{10}C_1(3) + {}^{10}C_2(-3)^2 + {}^{10}C_3(-3)^3 + \dots + {}^{10}C_9(-3)^9 + {}^{10}C_{10}(-3)^{10} \\ &= {}^{10}C_0 - {}^{10}C_1(3) + {}^{10}C_2(3)^2 - {}^{10}C_3(3)^3 + \dots + - {}^{10}C_9 3^9 + {}^{10}C_{10} 3^{10} \end{aligned} \quad \text{--- (3)}$$

$$\text{Now (2) + (3)}$$

$$\begin{aligned} 4^{10} + (-2)^{10} &= 2 {}^{10}C_0 + 2 {}^{10}C_2 3^2 + 2 {}^{10}C_4 3^4 + 2 {}^{10}C_6 3^6 + \dots + 2 {}^{10}C_{10} 3^{10} \\ 2^{20} + 2^{10} &= 2 ({}^{10}C_0 + {}^{10}C_2 3^2 + {}^{10}C_4 3^4 + \dots + {}^{10}C_{10} 3^{10}) \\ &\div 2 \end{aligned}$$

$$\begin{aligned} 2^{19} + 2^9 &= {}^{10}C_0 + {}^{10}C_2 3^2 + {}^{10}C_4 3^4 + \dots + {}^{10}C_{10} 3^{10} \\ \text{ie } 1 + {}^{10}C_2 3^2 + {}^{10}C_4 3^4 + \dots + {}^{10}C_{10} 3^{10} &= 2^9 (2^{10} + 1) \end{aligned}$$

$$\text{as } {}^{10}C_0 = 1.$$